

CISCO certified support technician -Cyber security



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In association with CYNUX ERA

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SRN : R25DE035

CLASS : 2ND SEM A SEC MCA

TITLE : CTF DEVELOPMENT

Objective

The primary objective of this project is to design, configure, and deploy a vulnerable Linux-based Capture The Flag (CTF) machine that provides a realistic environment for learning and practicing penetration testing techniques. The machine is intentionally configured with controlled vulnerabilities that allow participants to safely explore various stages of ethical hacking while improving their cybersecurity skills.

The CTF machine simulates a real-world networked system in which an attacker must identify and exploit security weaknesses to gain access to different parts of the system. Participants are required to perform a structured penetration testing methodology, beginning with information gathering and progressing through multiple stages until they successfully obtain all hidden flags. This approach helps users understand how attackers exploit vulnerabilities and how system administrators can implement effective security measures to prevent such attacks.

The project is designed to provide hands-on experience in the following areas:

- **Reconnaissance:** Discovering the target machine on the network and collecting publicly available information about the system using tools such as Nmap.
- **Enumeration:** Identifying running services, open ports, hidden directories, user accounts, and exposed files to understand the attack surface of the target machine.
- **Web Application Analysis:** Examining the hosted portfolio website for hidden clues, HTML comments, encoded credentials, and other information that may assist in further exploitation.
- **Credential Discovery:** Recovering hidden usernames and passwords through techniques such as Base64 decoding, source code inspection, and directory enumeration.
- **Steganography:** Extracting hidden files embedded within images using steganography tools, demonstrating how information can be concealed inside

digital media.

- **System Access:** Using the recovered credentials to establish secure remote access to the target system through the SSH service.
- **Privilege Escalation:** Identifying system misconfigurations or vulnerabilities that allow a low-privileged user to obtain administrative (root) access.
- **Flag Collection:** Locating hidden flag files distributed across different parts of the system, including the web server, user directories, embedded files, and the root directory.

Machine Specifications

Component Specification

Virtualization

Software

: VMware Workstation / VirtualBox

RAM : 16 GB

CPU : 2 vCPUs

Storage : 512 GB

Network Adapter : NAT

Operating System Used:

Target Machine

- **West-wild server**

Attacker Machine

- **Kali Linux 2025**

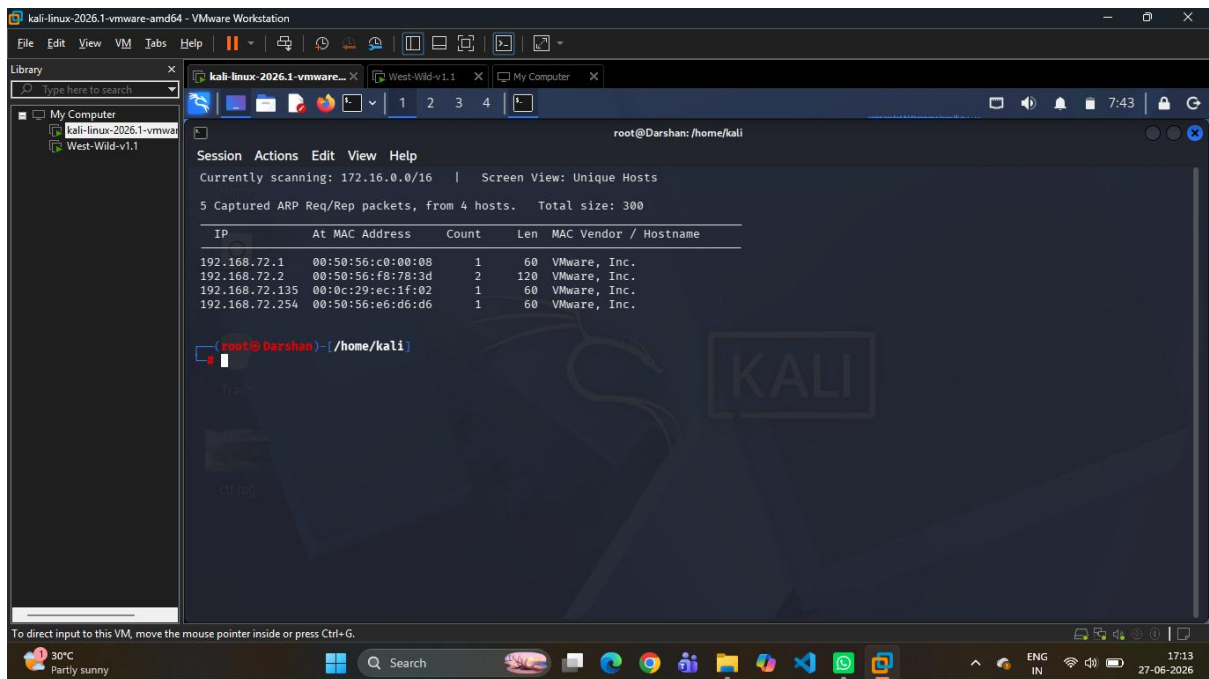
Network Configuration: Both virtual machines were configured in the same virtual network to enable communication.

Target Machine

- **Server : West-wild**

- IP Address: 192.168.72.135

Screenshot:

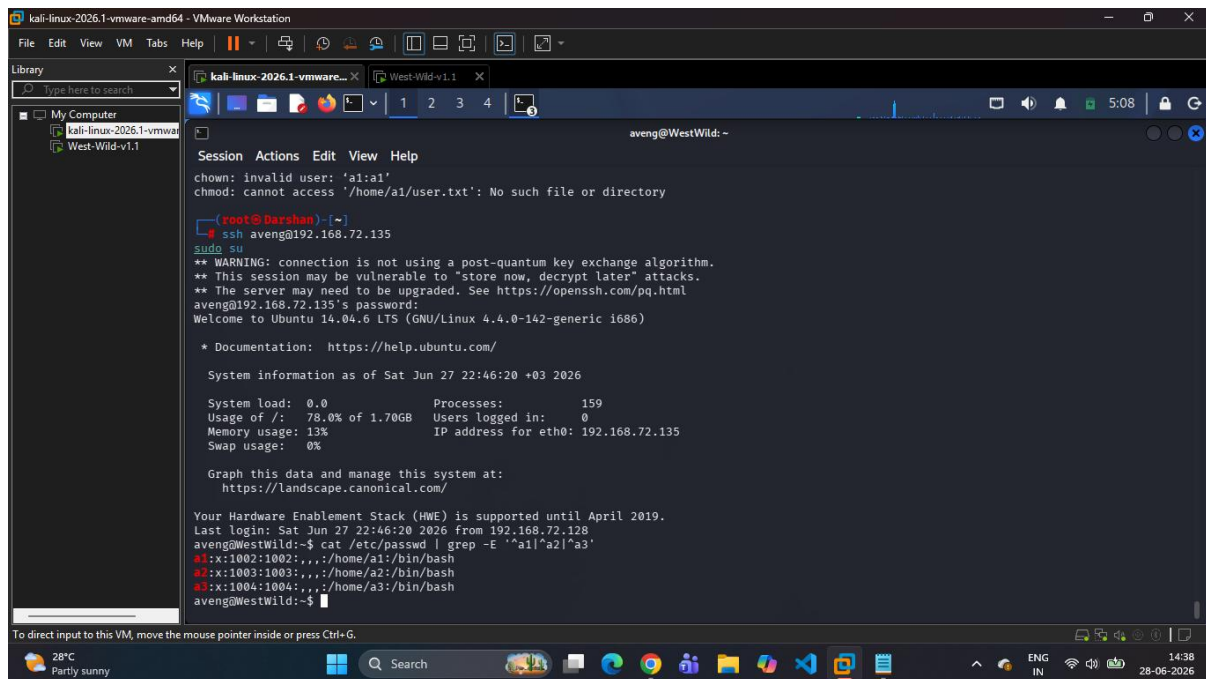


User Accounts Created

Three user accounts were created.

- 1.User – one
- 2.User – two
- 3.User – three

Screenshot:



Services Configured

The following services were configured:

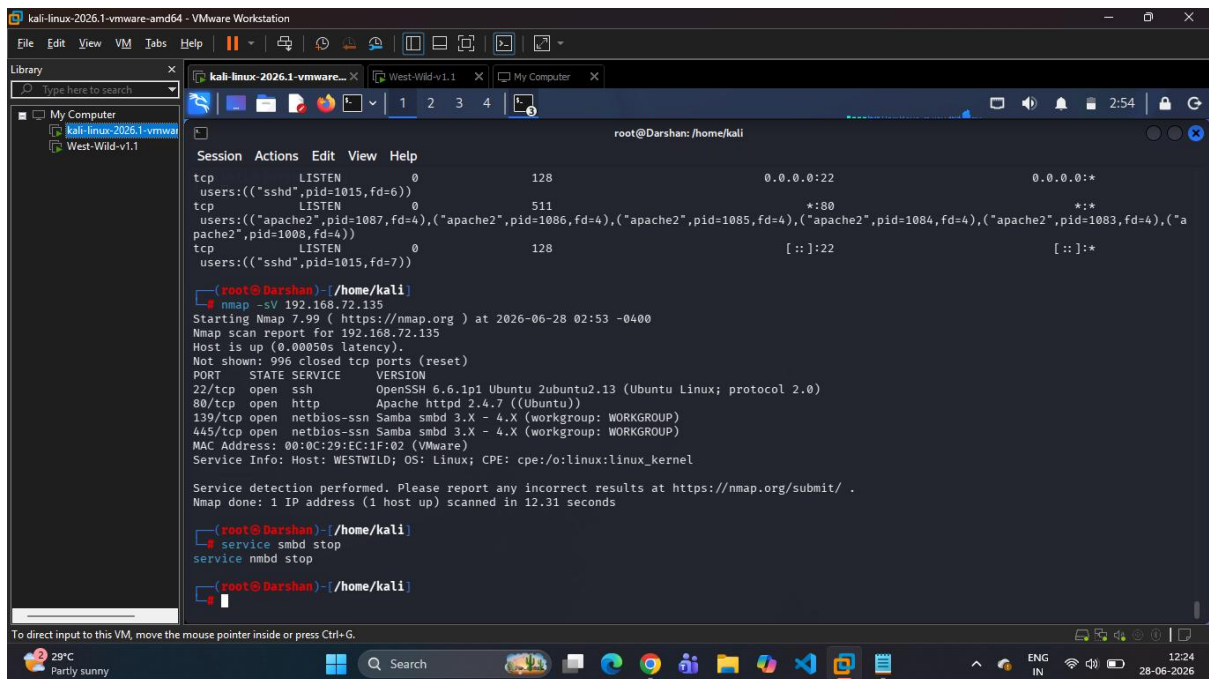
- SSH (Port 22)
- Apache Web Server (Port 80)

Firewall configuration allows only:

- Port 22
- Port 80

All other ports were blocked using UFW.

Ports available



```
root@Darshan: /home/kali
Session Actions Edit View Help
tcp LISTEN 0 128 0.0.0.0:22 0.0.0.0:*
users:(("sshd",pid=1015,fd=6))
tcp LISTEN 0 511 *:80 *:80
users:(("apache2",pid=1087,fd=4),("apache2",pid=1086,fd=4),("apache2",pid=1085,fd=4),("apache2",pid=1084,fd=4),("a
pache2",pid=1008,fd=4))
tcp LISTEN 0 128 [::]:22 [::]:*
users:(("sshd",pid=1015,fd=7))

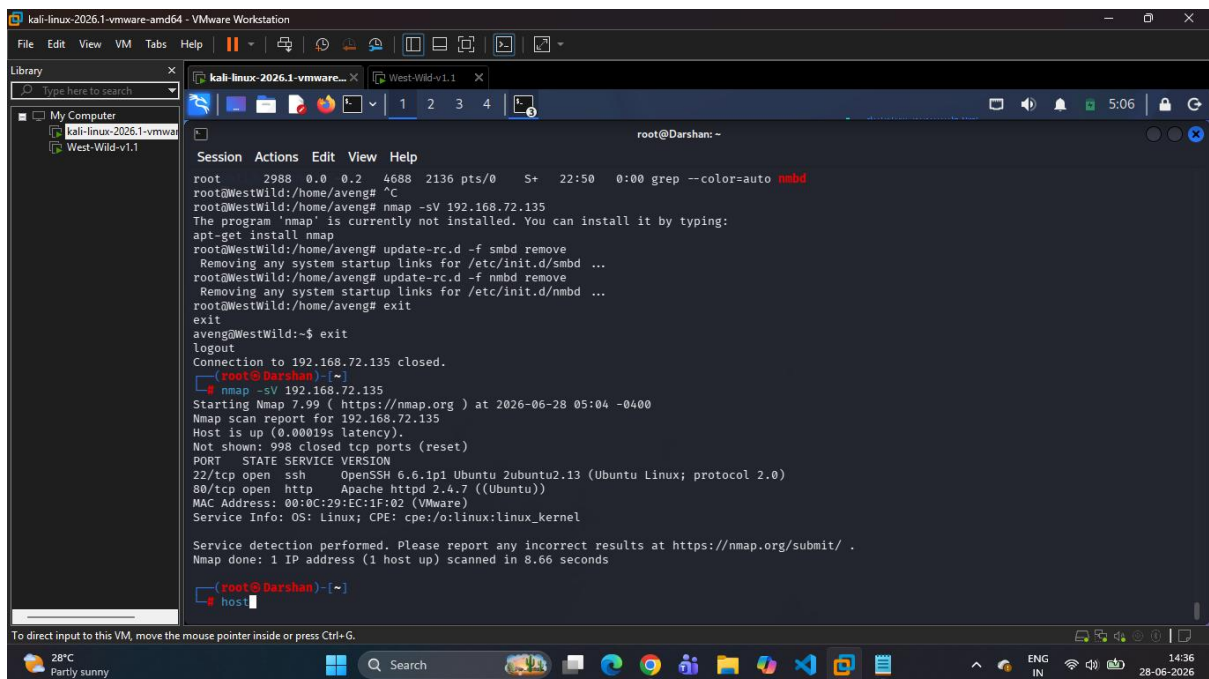
root@Darshan: /home/kali
nmap -sV 192.168.72.135
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-28 02:53 -0400
Nmap scan report for 192.168.72.135
Host is up (0.00050s latency).
Not shown: 996 closed tcp ports (reset)
PORT STATE SERVICE VERSION
22/tcp open ssh OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp open http Apache httpd 2.4.7 ((Ubuntu))
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
MAC Address: 00:0C:29:EC:1F:02 (VMware)
Service Info: Host: WESTWILD; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.31 seconds

root@Darshan: /home/kali
service smb stop
service nmbd stop

root@Darshan: /home/kali
```

After removed



```
root@Darshan: ~
Session Actions Edit View Help
root 2988 0.0 0.2 4688 2136 pts/0 S+ 22:50 0:00 grep --color=auto nmbd
root@WestWild:/home/aveng# ^C
root@WestWild:/home/aveng# nmap -sV 192.168.72.135
The program 'nmap' is currently not installed. You can install it by typing:
apt-get install nmap
root@WestWild:/home/aveng# update-rc.d -f smbd remove
Removing any system startup links for /etc/init.d/smbd ...
root@WestWild:/home/aveng# update-rc.d -f nmbd remove
Removing any system startup links for /etc/init.d/nmbd ...
root@WestWild:/home/aveng# exit
exit
aveng@WestWild:~$ exit
logout
Connection to 192.168.72.135 closed.

root@Darshan: ~
nmap -sV 192.168.72.135
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-28 05:04 -0400
Nmap scan report for 192.168.72.135
Host is up (0.00019s latency).
Not shown: 998 closed tcp ports (reset)
PORT STATE SERVICE VERSION
22/tcp open ssh OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp open http Apache httpd 2.4.7 ((Ubuntu))
MAC Address: 00:0C:29:EC:1F:02 (VMware)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.66 seconds

root@Darshan: ~
host
```

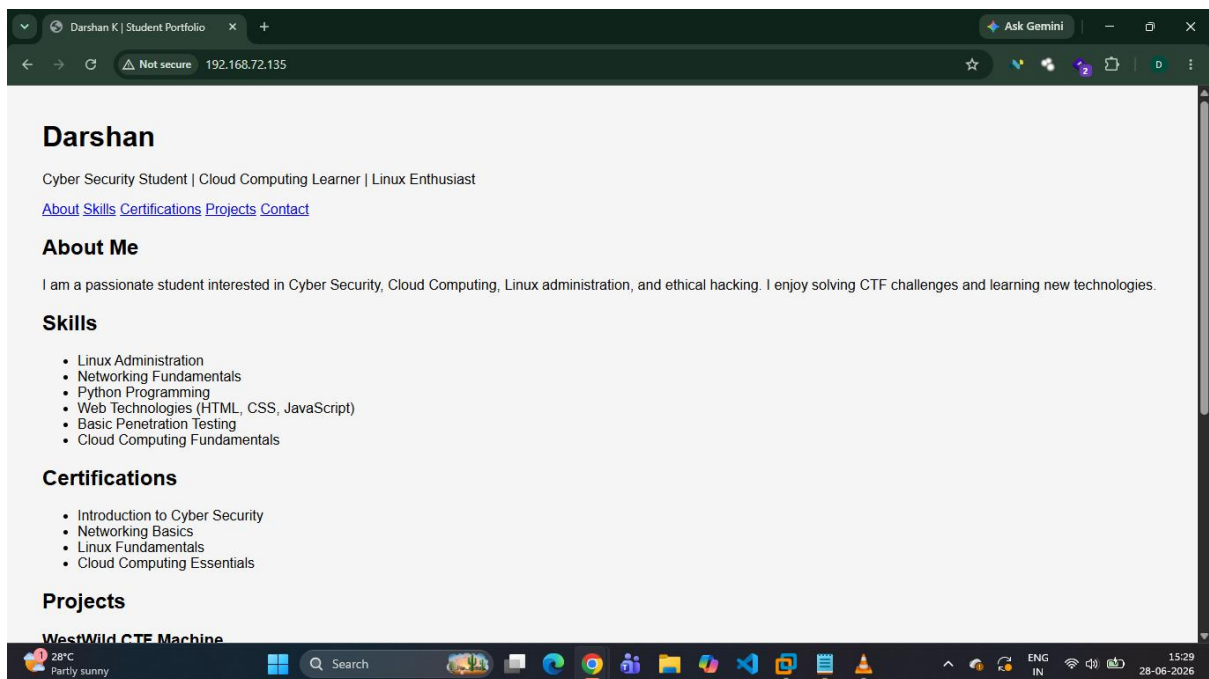
Website Design Overview

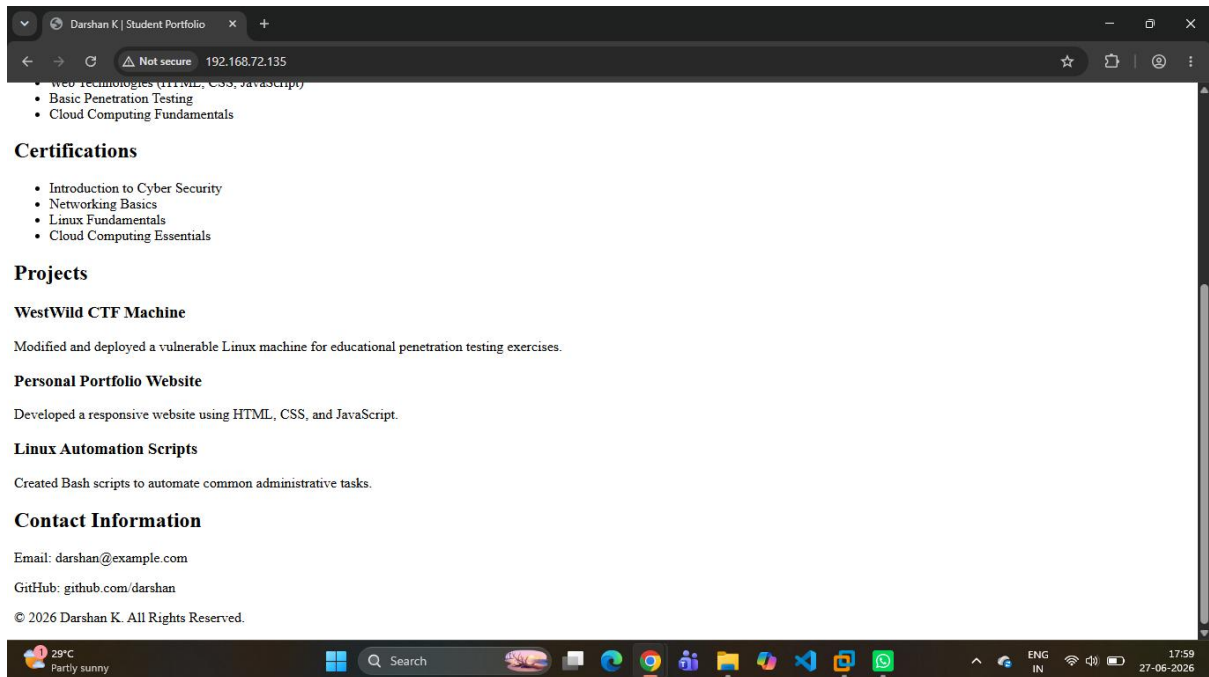
A portfolio website was developed and hosted using Apache.

The website includes:

- Professional Profile
- Skills
- Certifications
- Projects
- Contact Information

The website was intentionally designed to contain hidden clues for CTF participants.





Hidden Credentials Methodology:

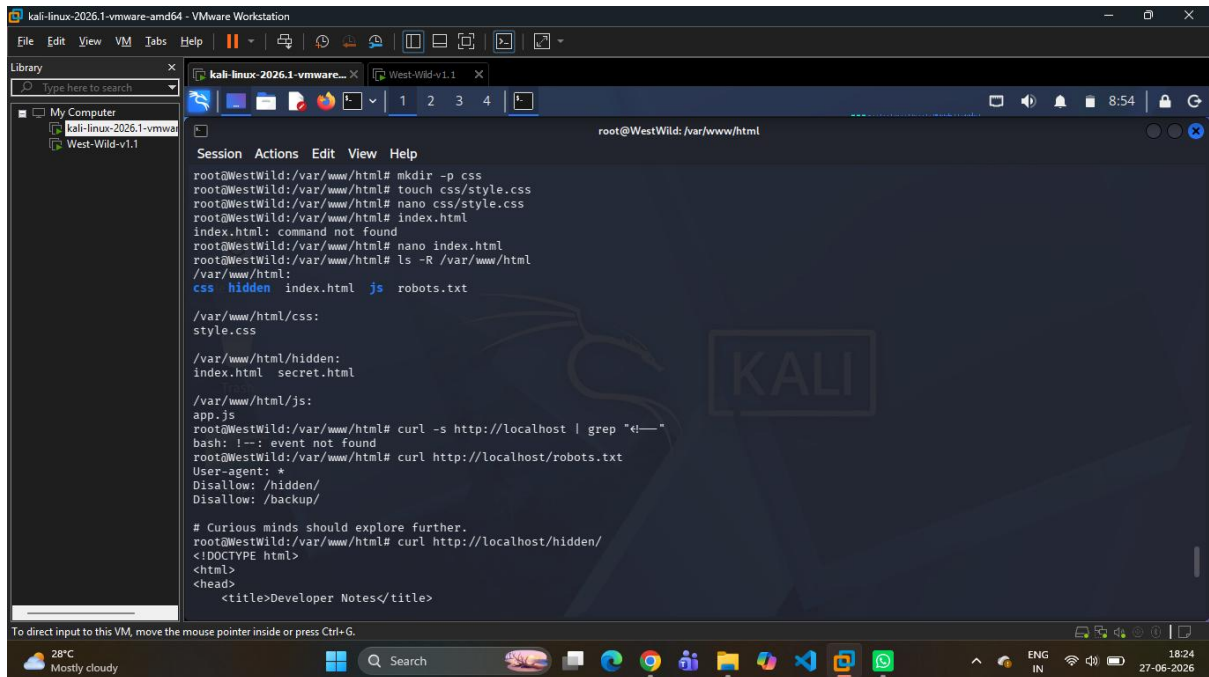
To increase the difficulty of the challenge, credentials were hidden using multiple techniques.

Methods include:

- Base64 encoded username/password
- Hidden HTML comments
- Source code comments
- Hidden files
- Directory enumeration

Participants must inspect the website source code and hidden files to discover login

credentials.



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window is titled 'root@WestWild: /var/www/html'. The user has created a directory 'css' and a file 'style.css'. They have also created a file 'index.html' and a file 'secret.html' in the 'hidden' directory. The terminal output shows the following commands and results:

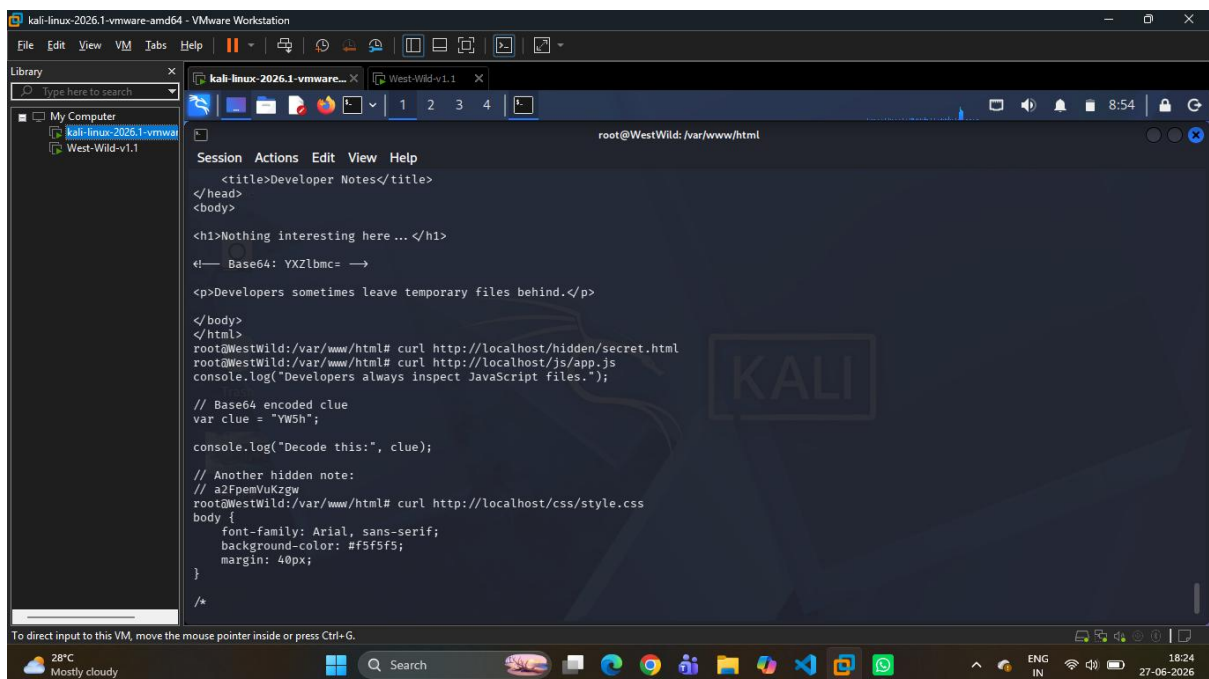
```
root@WestWild:/var/www/html# mkdir -p css
root@WestWild:/var/www/html# touch css/style.css
root@WestWild:/var/www/html# nano css/style.css
root@WestWild:/var/www/html# index.html
index.html: command not found
root@WestWild:/var/www/html# nano index.html
root@WestWild:/var/www/html# ls -R /var/www/html
/var/www/html:
css  hidden  index.html  js  robots.txt

/var/www/html/css:
style.css

/var/www/html/hidden:
index.html  secret.html

/var/www/html/js:
app.js
root@WestWild:/var/www/html# curl -s http://localhost | grep "e—"
bash: !-: event not found
root@WestWild:/var/www/html# curl http://localhost/robots.txt
User-agent: *
Disallow: /hidden/
Disallow: /backup/

# Curious minds should explore further.
root@WestWild:/var/www/html# curl http://localhost/hidden/
<!DOCTYPE html>
<html>
<head>
<title>Developer Notes</title>
```



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window is titled 'root@WestWild: /var/www/html'. The user has created a directory 'css' and a file 'style.css'. They have also created a file 'index.html' and a file 'secret.html' in the 'hidden' directory. The terminal output shows the following commands and results:

```
root@WestWild:/var/www/html# curl http://localhost/hidden/secret.html
root@WestWild:/var/www/html# curl http://localhost/js/app.js
console.log("Developers always inspect JavaScript files.");

// Base64 encoded clue
var clue = "YW5h";

console.log("Decode this:", clue);

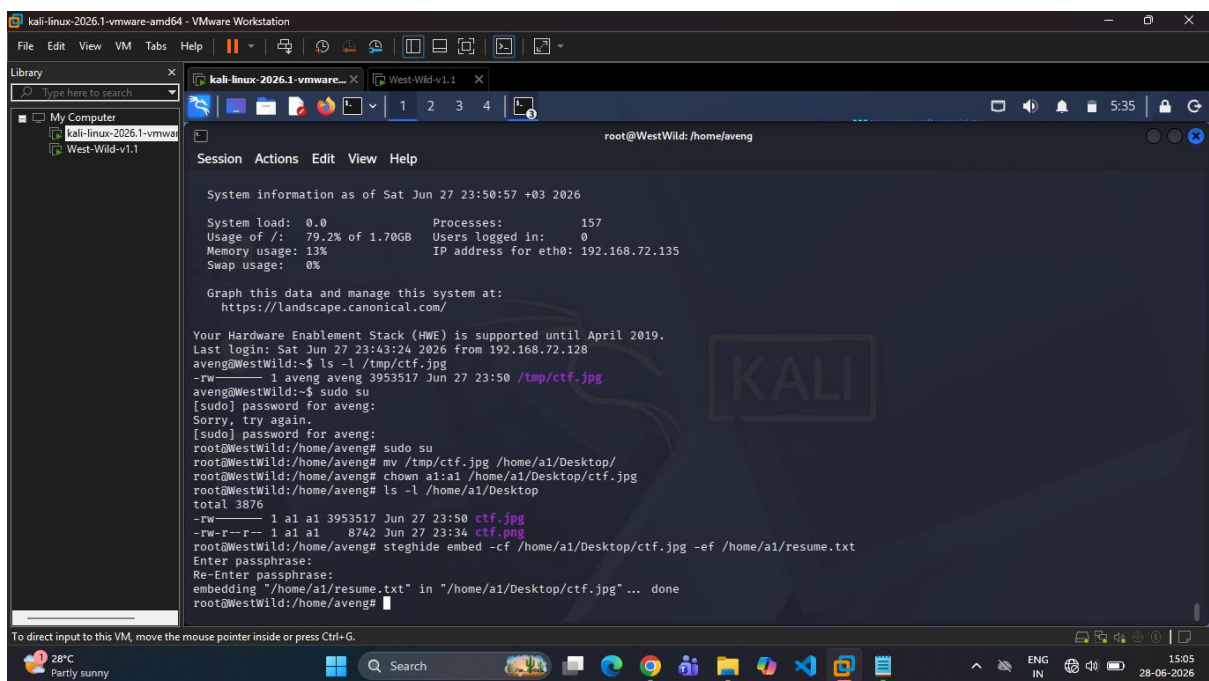
// Another hidden note:
// a2FpemVukZgw
root@WestWild:/var/www/html# curl http://localhost/css/style.css
body {
  font-family: Arial, sans-serif;
  background-color: #f5f5f5;
  margin: 40px;
}
```

Steganography Implementation

A resume file was embedded inside an image using Steghide.

Implementation Steps:

1. Resume prepared.
2. Cover image selected.
3. Resume embedded using:



```
kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
kali-linux-2026.1-vmware-amd64
West-Wild-v1.1

root@WestWild: /home/aveng

Session Actions Edit View Help

System information as of Sat Jun 27 23:50:57 +03 2026

System load: 0.0      Processes:      157
Usage of /:  79.2% of 1.70GB   Users logged in: 0
Memory usage: 13%      IP address for eth0: 192.168.72.135
Swap usage:  0%

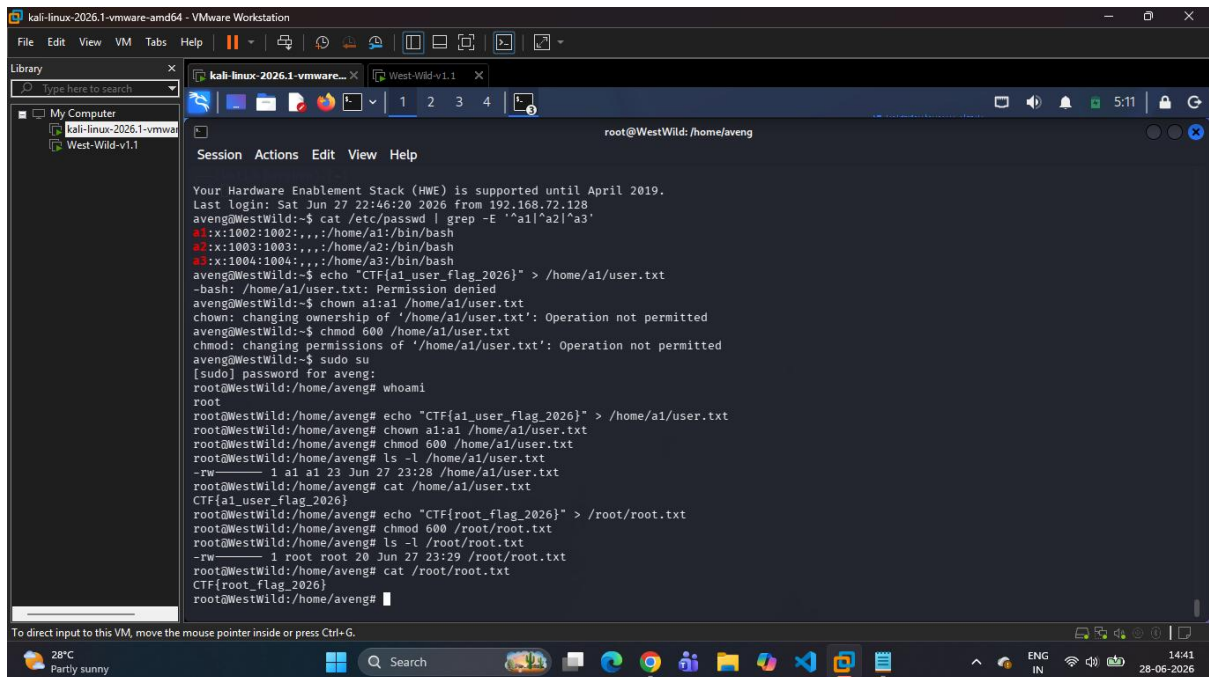
Graph this data and manage this system at:
https://landscape.canonical.com/

Your Hardware Enablement Stack (HWE) is supported until April 2019.
Last login: Sat Jun 27 23:43:24 2026 from 192.168.72.128
aveng@WestWild:~$ ls -l /tmp/ctf.jpg
-rw-r--r-- 1 aveng aveng 3953517 Jun 27 23:50 /tmp/ctf.jpg
aveng@WestWild:~$ sudo su
[sudo] password for aveng:
Sorry, try again.
[sudo] password for aveng:
root@WestWild:~# sudo su
root@WestWild:~# mv /tmp/ctf.jpg /home/a1/Desktop/
root@WestWild:~# chown a1:a1 /home/a1/Desktop/ctf.jpg
root@WestWild:~# ls -l /home/a1/Desktop
total 3876
-rw-r--r-- 1 a1 a1 3953517 Jun 27 23:50 ctf.jpg
-rw-r--r-- 1 a1 a1 8742 Jun 27 23:34 ctf.png
root@WestWild:~# steghide embed -cf /home/a1/Desktop/ctf.jpg -ef /home/a1/resume.txt
Enter passphrase:
Re-Enter passphrase:
Embedding "/home/a1/resume.txt" in "/home/a1/Desktop/ctf.jpg" ... done
root@WestWild:~#
```

Flag Locations

In user

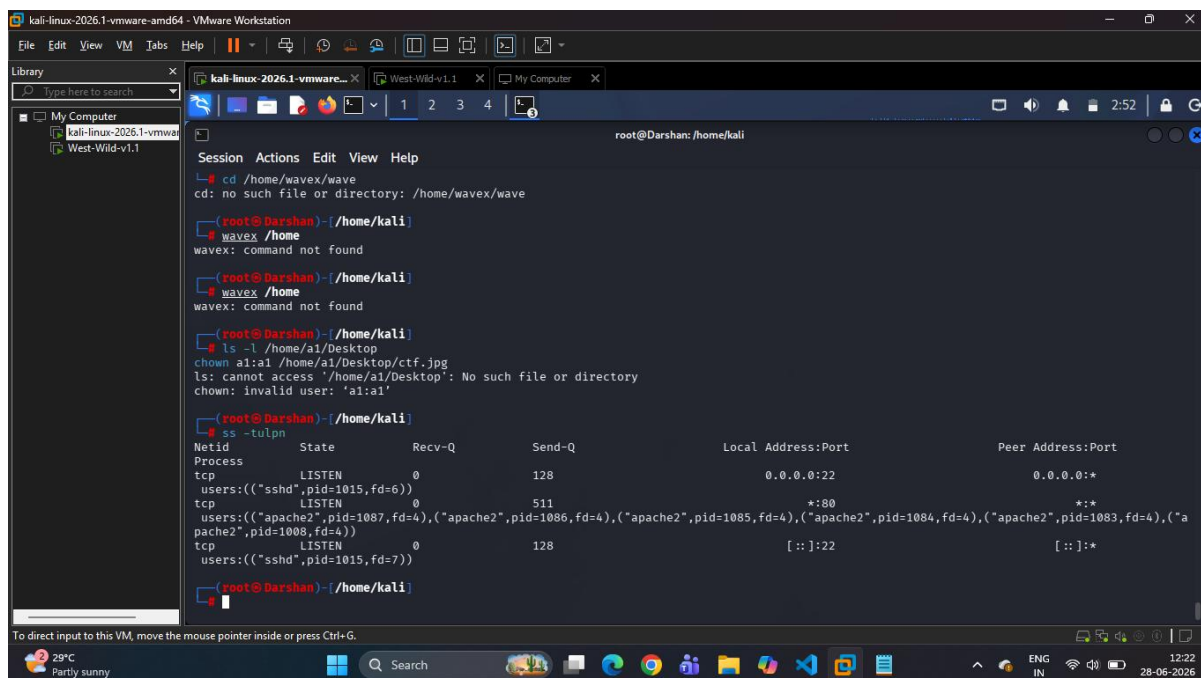
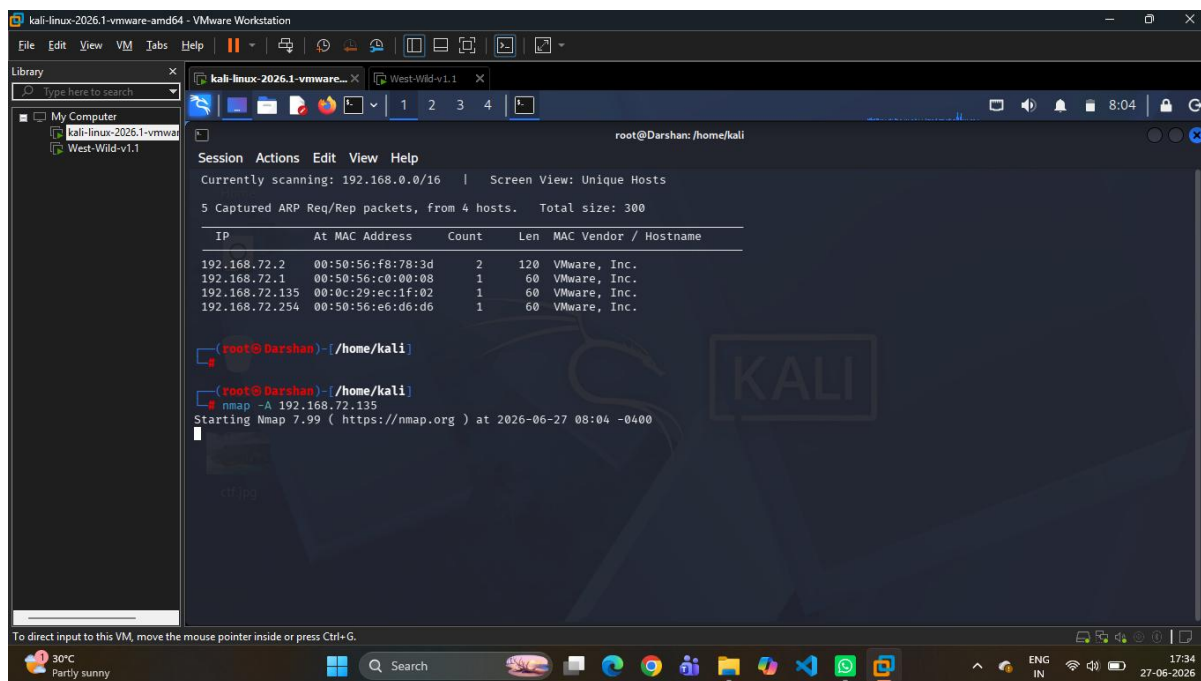
In root

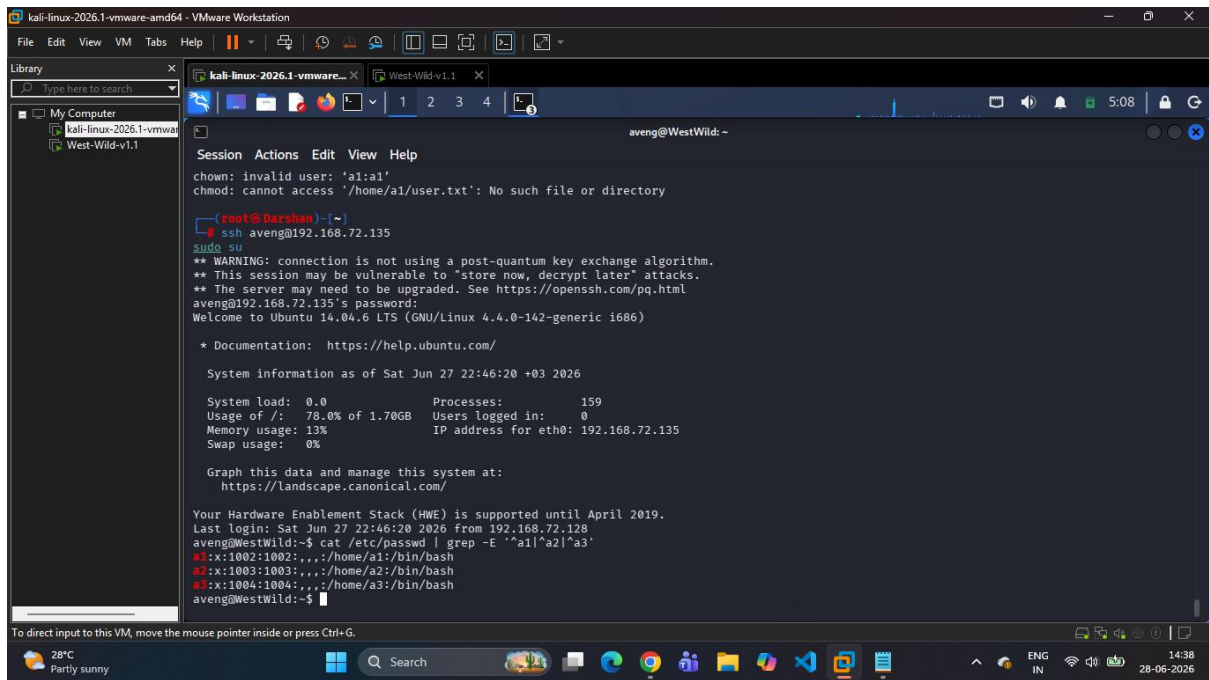


The screenshot shows a Kali Linux virtual machine running on VMware Workstation. The terminal window displays the following commands and output:

```
root@WestWild: /home/aveng
Your Hardware Enablement Stack (HWE) is supported until April 2019.
Last login: Sat Jun 27 22:46:20 2026 from 192.168.72.128
aveng@WestWild:~$ cat /etc/passwd | grep -E '^a1|^a2|^a3'
a1:x:1002:1002:,,,:/home/a1:/bin/bash
a2:x:1003:1003:,,,:/home/a2:/bin/bash
a3:x:1004:1004:,,,:/home/a3:/bin/bash
aveng@WestWild:~$ echo "CTF{a1_user_flag_2026}" > /home/a1/user.txt
-bash: /home/a1/user.txt: Permission denied
aveng@WestWild:~$ chown a1:a1 /home/a1/user.txt
chown: changing ownership of '/home/a1/user.txt': Operation not permitted
aveng@WestWild:~$ chmod 600 /home/a1/user.txt
chmod: changing permissions of '/home/a1/user.txt': Operation not permitted
aveng@WestWild:~$ sudo su
[sudo] password for aveng:
root@WestWild: /home/aveng# whoami
root
root@WestWild: /home/aveng# echo "CTF{a1_user_flag_2026}" > /home/a1/user.txt
root@WestWild: /home/aveng# chown a1:a1 /home/a1/user.txt
root@WestWild: /home/aveng# chmod 600 /home/a1/user.txt
root@WestWild: /home/aveng# ls -l /home/a1/user.txt
-rw-r--r-- 1 a1 a1 23 Jun 27 23:28 /home/a1/user.txt
root@WestWild: /home/aveng# cat /home/a1/user.txt
CTF{a1_user_flag_2026}
root@WestWild: /home/aveng# echo "CTF{root_flag_2026}" > /root/root.txt
root@WestWild: /home/aveng# chmod 600 /root/root.txt
root@WestWild: /home/aveng# ls -l /root/root.txt
-rw-r--r-- 1 root root 20 Jun 27 23:29 /root/root.txt
root@WestWild: /home/aveng# cat /root/root.txt
CTF{root_flag_2026}
root@WestWild: /home/aveng#
```

Screenshots of Configuration





Testing Procedures

The developed CTF machine was tested from an attacker machine running Kali Linux to ensure that all intended services, vulnerabilities, and challenges functioned correctly.

1. Network Connectivity Testing

- Verified network communication using ping.
- Confirmed that the target machine was reachable from the attacker machine.
- Ensured the VM used the correct NAT network configuration.

2. Port Scanning

- Performed service enumeration using Nmap:

```
nmap -sV <target-ip>
```

- Verified that only the required ports were accessible:
 - SSH (22/tcp)
 - HTTP (80/tcp)
- Disabled unnecessary Samba services (139/tcp and 445/tcp).

3. Web Enumeration Testing

The web application was tested for hidden resources through multiple enumeration techniques:

- Accessed `robots.txt`.
- Inspected HTML source code comments.
- Examined JavaScript files for encoded strings.
- Reviewed CSS files for hidden clues.
- Explored hidden directories such as `/hidden/`.

Commands used:

```
curl http://<target-ip>/robots.txt
curl http://<target-ip>/hidden/
curl http://<target-ip>/js/app.js
curl http://<target-ip>/css/style.css
```

4. Credential Discovery Validation

Base64-encoded clues were successfully decoded to reveal usernames and password fragments.

Examples:

```
echo "YW5h" | base64 -d
echo "a2FpemVuKzgw" | base64 -d
```

5. User Flag Testing

User-level access was validated by retrieving:

```
cat /home/al/user.txt
```

The user flag was successfully obtained after discovering valid credentials.

6. Root Privilege Testing

Privilege escalation paths were tested to ensure that the root flag could only be accessed after obtaining administrative privileges.

```
cat /root/root.txt
```

7. Steganography Verification

Steghide was used to embed hidden information inside a JPEG image.

Verification command:

```
steghide info ctf.jpg
```

The embedded file extraction process was tested using the correct passphrase.

8. Service Validation

The following services were confirmed operational:

- OpenSSH Server (Port 22)
- Apache Web Server (Port 80)

All unnecessary services were disabled to minimize the attack surface.

Identified Vulnerabilities

The CTF machine intentionally includes several vulnerabilities for educational purposes.

1. Information Disclosure Through robots.txt

The `robots.txt` file exposes hidden directories that should not be publicly indexed.

Impact:

- Enables attackers to discover sensitive locations.

2. Hidden HTML Comments

Comments within web pages reveal clues and hints.

Impact:

- Attackers inspecting source code can gather useful intelligence.

3. Base64-Encoded Credentials

Username and password fragments are encoded using Base64.

Impact:

- Weak obfuscation methods can be easily reversed.

4. Sensitive Information in JavaScript Files

JavaScript files contain hidden variables and encoded data.

Impact:

- Client-side code should never store secrets.

5. Steganography-Based Data Hiding

Important information is concealed inside image files.

Impact:

- Attackers must perform forensic analysis to discover hidden content.

6. Weak Credential Management

Password hints are distributed across multiple files.

Impact:

- Attackers can combine clues from different sources to gain access.

7. Privilege Escalation Opportunities

Misconfigurations and accessible information may allow escalation from user privileges to root access.

Impact:

- Full system compromise becomes possible.

These vulnerabilities are intentionally implemented to provide realistic penetration-testing scenarios and improve practical cybersecurity skills.

Conclusion

The CTF machine successfully simulates a real-world penetration testing environment by incorporating multiple stages of reconnaissance, enumeration, exploitation, and privilege escalation.

Participants are required to use common cybersecurity tools such as Nmap, Curl, SSH, Base64 decoders, and Steghide to discover hidden information and capture both user and root flags. The machine demonstrates how seemingly minor information disclosures, hidden files, weak credential practices, and steganography techniques can contribute to a complete system compromise.

Throughout the development process, unnecessary services were removed, required services were validated, and testing procedures confirmed that the intended challenge path functions correctly. The project provides valuable hands-on experience in web enumeration, Linux administration, privilege escalation, and digital forensics.

Overall, the CTF machine meets the assignment objectives and offers an educational platform for developing practical offensive security skills in a controlled environment.